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UW12GMultiToneSrcDefTopic



About the UltraWave 12G MWMultiToneSource

[About the UltraWave 12G MWMultiToneSource](#)

The MWMultiToneSource is a programmable RF source that generates two-tone or multitone signals in the frequency range from 380MHz to 12GHz. Applications for the MWMultiToneSource

include IMD, IP2, and IP3 tests.

This section includes the following topics:

• [Features](#)

• [Safety Information](#)

• [Key Terms](#)

MWMultiToneSource information includes the following sections:

• [UltraWave 12G MWMultiToneSource Theory of Operation](#)

• [UltraWave 12G MWMultiToneSource Programming](#)

• [UltraWave 12G MWMultiToneSource Debug Display](#)

Related Topics

• [UltraWave 12G Specifications](#)

• [MWMultiToneSource \(VBT syntax\)](#)

• [UltraWave 12G DIB Design Guidelines](#)

• [About the UltraWave 12G Subsystem](#)

• [UltraWave 12G Examples](#)

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About the UltraWave 12G MWMultiToneSource : Features

Features

The MWMultiToneSource generates either two-tone signals by adding two CW signals, or more complex signals by adding a modulated signal to a CW signal or by adding two modulated signals. Basic functional specifications are:

- | | |
|---|---|
| • | Added signals components can be CW or modulated |
|---|---|
- | | |
|---|--|
| • | CW signal frequency range: 380MHz to 12GHz |
|---|--|
- | | |
|---|---|
| • | Modulated signal frequency range: 76MHz to 6GHz |
|---|---|
- | | |
|---|---------------------|
| • | CW amplitude range: |
|---|---------------------|
- | | |
|---|--|
| – | Primary signal path: -130dBm to +13dBm |
|---|--|
- | | |
|---|---|
| – | Secondary signal path: -100dBm to -7dBm |
|---|---|

For UltraWave 12G hardware specifications, see [UltraWave 12G Specifications](#).

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About the UltraWave 12G MWMultiToneSource : Safety Information

Safety Information

When operating the Microwave instruments, always follow general UltraFLEX test system safety guidelines. See Test System Safety.

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About the UltraWave 12G MWMultiToneSource : Key Terms

Key Terms

add path	Signal combiner allowing two RF signals to be added to create a two-tone signal or a more complex multitone signal if one or both component signals is modulated.
multitone	A signal composed of two or more discrete frequencies. For the MWMultiToneSource, a multitone is created by adding two signals, each of which can be a CW or modulated signal.

For more MW terms and definitions, see [Key Terms](#) in About the UltraWave 12G Subsystem.

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UltraWave 12G MWMultiToneSource Debug Display

UltraWave 12G MWMultiToneSource Debug Display

The UltraWave 12G Microwave Debug Display can display one or more instrument blocks for the MWMultiToneSource logical instrument, each showing the state of a MWMultiToneSource in the context of a site.

This section includes the following topic:

- [UltraWave 12G MWMultiToneSource Debug Display Controls](#)

For an overview of all displays supporting the UltraWave 12G subsystem, see [UltraWave 12G Displays](#).

For more information on using debug displays, see the following:

- [Using Debug Displays in TDE](#)

- [Producing Code from Debug Displays](#)

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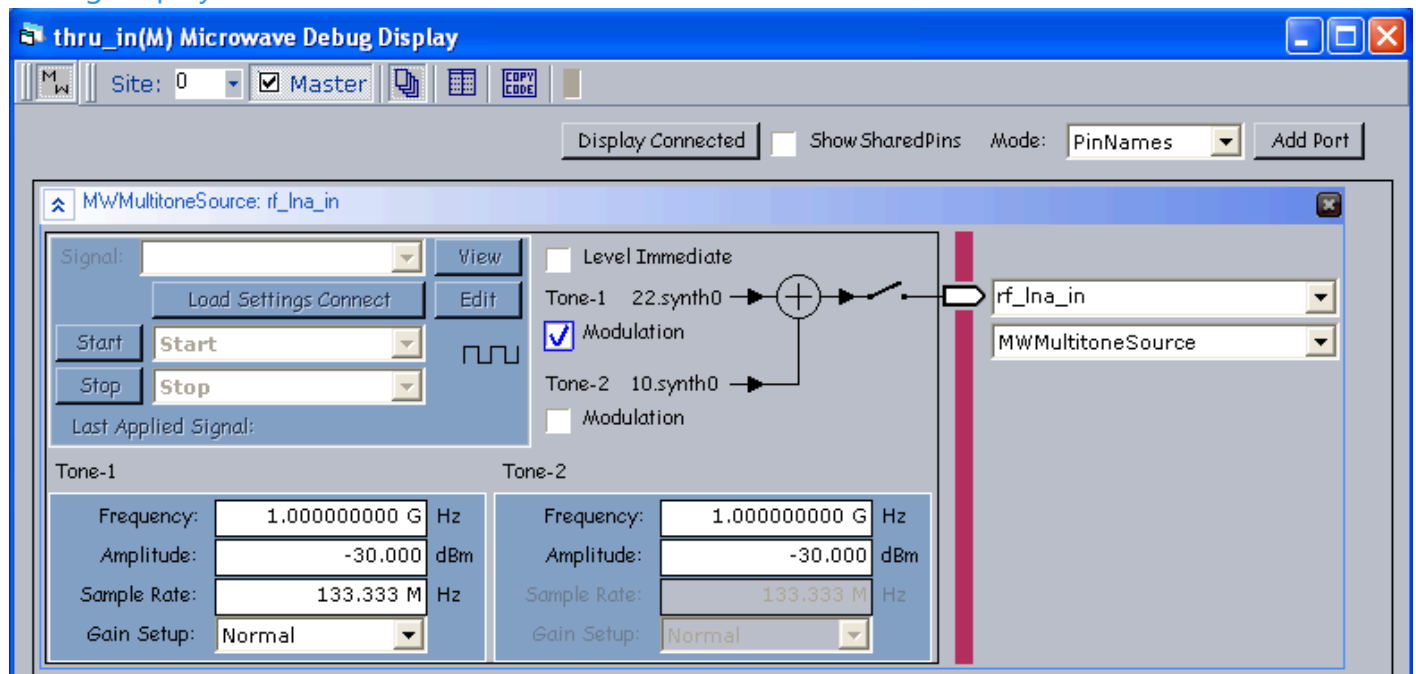


[UltraWave 12G MWMultiToneSource Debug Display](#) : UltraWave 12G MWMultiToneSource Debug Display Controls

UltraWave 12G MWMultiToneSource Debug Display Controls

The display controls for the MWMultiToneSource assigned to a port allow you to view and set instrument parameters, which are also accessible through the Pins interface of the MWMultiToneSource VBT language.

For information about the common debug display controls, see [UltraWave12G Microwave Board Debug Display](#).



MWMultiToneSource Logical Instrument Display

Tone1 always controls the primary signal path. For the example display above, Tone1 is a modulated signal and Tone2 is a CW signal.

The following controls are available on the MWMultiToneSource logical instrument block:

Frequency	Displays and sets the signal frequency. The display has separate frequencies for Tone1 and Tone2. The tooltip displays the allowable frequency range.
Amplitude	Displays and sets the signal amplitude. The display has separate amplitudes for Tone1 and Tone2. The tooltip displays the allowable amplitude range.
Level Immediate	Sets the MWMultiToneSource leveling mode to Level Immediate, when checked. See MWMultiToneSource.Pins.LevelImmediate in the VBT Syntax Reference.
Modulation	Sets the tone to use a modulated signal, when checked.
Sample Rate	Displays and sets the modulator sample rate for a modulated tone. Enabled only if Modulation is checked. The tooltip displays the allowable sample rate range.
Gain Setup	Displays and sets the gain setup for a modulated tone on a drop-down list: • Normal • LowDistortion Enabled only if Modulation is checked.
synth label	Identifies the two synthesizers (slot and number) used by this MWMultiToneSource logical instrument for Tone1 (primary) and Tone2.
Signal	Displays the selected source Signal on a drop-down list of available signals.
Load Settings Connect	Loads the selected signal and connects the MWMultiToneSource.
View	Displays a list of signal viewer tools (such as WaveScope or VectorScope) for the selected Signal on a drop-down list of available viewers.
Info	Displays information for the selected Signal in a pop-up balloon window.
Start	Starts the selected Signal using one of the start options: • Start

- [Start1](#)
- [StartE](#)
- [Start1E](#)

Stop

[Stops the selected Signal using one of the stop options:](#)

- [Stop](#)
- [StopE](#)

Last Applied Signal
connection relay

[Displays the name of the Signal last loaded to the MWMultiToneSource.](#)
[The relay control connects or disconnects the MWMultiToneSource logical instrument to the specified pin or channel and displays the connection state. Click the relay to open or close the connection.](#)

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UltraWave 12G MWMultiToneSource Programming

UltraWave 12G MWMultiToneSource Programming

The MWMultiToneSource is a MW logical instrument in the IG-XL environment. In general, you program the MWMultiToneSource like the other MW instruments.

The following topics are available:

- Sourcing Two-Tone Signals Using the MWMultiToneSource
- MWMultiToneSource Pattern Control

Before writing a program using the MWMultiToneSource, see [UltraWave 12G Instrument Programming](#).

For concepts, a procedure, and an example, see [Sourcing Two-Tone Signals Using the MWMultiToneSource](#).

For more information, see [MWMultiToneSource](#) in the VBT Syntax Reference.

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UltraWave 12G MWMultiToneSource Programming : Sourcing Two-Tone Signals Using the MWMultiToneSource

Sourcing Two-Tone Signals Using the MWMultiToneSource
This section includes the following topics:

- | | |
|---|--------------------------------------|
| • | Programming Concepts |
| • | Programming Steps |
| • | Examples |

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UltraWave 12G MWMultiToneSource Programming : Sourcing Two-Tone Signals Using the MWMultiToneSource : Programming Concepts

Programming Concepts

Source Level Calibration

Source level calibration, also called leveling, finely calibrates the MWMultiToneSource at each frequency set in a program. Leveling takes place at run time.

Leveling occurs the first time a specific frequency is set for an individual tone. Later VBT statements setting the same frequency use cached data and do not level, unless the program invokes leveling using the Pins.LevelImmediate VBT property.

TSD File Support

The MWMultiToneSource accepts signal definitions for modulated signals in TSD files. TSD files are loaded using the Pins.Signals.Tones.SignalDefinitionFile property.

For more information, see Using the ESA Toolkit.

MWMultiToneSource Reset Values

The table lists the local reset values for the MWMultiToneSource.

MWMultiToneSource Reset Values

Setting	Reset Value
Amplitude (Tones(1))	-30dBm
Frequency (Tones(1))	1GHz
Modulation	0
Amplitude 2 (Tones(2))	-30dBm
Frequency 2 (Tones(2))	1GHz

Modulation	0
Connections	Disconnected



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UltraWave 12G MWMultiToneSource Programming : Sourcing Two-Tone Signals Using the MWMultiToneSource : Programming Steps

Programming Steps

The MWMultiToneSource can produce a two-tone signal (two CW signals), each with a programmed frequency and amplitude. The MWMultiToneSource can also produce a multitone by combining two modulated signals or by combining a CW signal with a modulated signal.

To source a two-tone signal to an assigned RF port:

1. Set source signal amplitude and frequency for tone 1 using the Tones interface.
2. Set source signal amplitude and frequency for tone 2 using the Tones interface.
3. Connect to the DUT pin or channel.

To source a multitone signal created by combining two modulated signals to an assigned RF port:

1. Add a new Signal to the MWMultiToneSource collection.
2. Set Signal parameters for tone 1 using the Tones interface.
3. Set Signal parameters for tone 2 using the Tones interface.
4. Turn on modulation for both Signals.
5. Connect to the DUT pin or channel using LoadSettingsConnect.

6. [Start the Signal or Signals using a VBT statement or by running a pattern.](#)

[See Examples.](#)

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UltraWave 12G MWMultiToneSource Programming : Sourcing Two-Tone Signals Using the MWMultiToneSource : Examples

Examples

The following examples are available:

- [Two-Tone Signal](#)
- [Multitone Signal from Two Modulated Signals](#)

Two-Tone Signal

This example code function accepts two separate frequencies (tones) and a single amplitude (RF power) for both.

```
Public Function MWMultiToneSourceSignal(ByVal Source_Pin As String, _
    ByVal F1_freq As Double, _
    ByVal F2_freq As Double, _
    ByVal RF_power As Double) As Long
```

```
Dim F1_power_level As Double
```

```
Dim F2_power_level As Double
```

```
' Set up low-side tone F1 using MW source1
```

```
F1_power_level = RF_power
```

```
With TheHdw.MWMultiToneSource.Pins(Source_Pin).Tones(1)
    .Frequency = F1_freq
```

```
.Amplitude = F1_power_level
```

```
End With
```

```
' Set up high side tone F2 using MW source2
```

```
F2_power_level = RF_power
```

```
With TheHdw.MWMultiToneSource.Pins(Source_Pin).Tones(2)
```

```
.Frequency = F2_freq
```

```
.Amplitude = F2_power_level
```

```
End With
```

```
' Connect the multitone source
```

```
TheHdw.MWMultiToneSource.Pins(Source_Pin).Connect
```

```
End Function
```

Multitone Signal from Two Modulated Signals

This example code function uses Gen4 waveforms to create a multitone signal. The function accepts frequency, amplitude (level), waveform file, IF frequency (required for Gen4 waveforms), and other Signal parameters for each modulated "tone."

```
Public Function SourceMultiTone(ByVal rf_freq As Double, _
```

```
    ByVal rf_freq1 As Double, _
```

```
    ByVal rf_level As Double, _
```

```
    ByVal rf_level1 As Double, _
```

```
    ByVal modsrc_segname As String, _
```

```
    ByVal modsrc_segname1 As String, _
```

```
    ByVal SigName As String
```

```
    ByVal IfFreqOne As Double, _
```

```
    ByVal IfFreqTwo As Double, _
```

```
    ByVal sampleRateOne As Double, _
```

```
    ByVal sampleRateTwo As Double, _
```

```
    ByVal sampleSzOne As Long, _
```

```
    ByVal sampleSzTwo As Long) As Long
```

```
' Add a new Signal to the MWMultiToneSource Signals collection and set
```

```
' all Signal parameters to default values.
```

```
TheHdw.MWMultiToneSource.Pins("RF_PIN").Signals.Add (SigName)
```

```
' Set the name of the Signal used by default when a Signal is applied
```

' without a Signal name.

```
TheHdw.MWMultiToneSource.Pins("RF_PIN").Signals.DefaultSignal = SigName
```

' If using Gen4 waveforms, IF Frequency must be specified.

' The UltraWave 12G MWMultiToneSource uses IFFrequency as input parameters

' to convert the IF data to IQ data before loading it into the AWG.

```
TheHdw.MWMultiToneSource.Pins("RF_PIN").Signals(SigName) _  
    .Tones(1).IFFrequency = IfFreqOne
```

```
TheHdw.MWMultiToneSource.Pins("RF_PIN").Signals(SigName).Tones(2). _  
    IFFrequency = IfFreqTwo
```

' Set up parameters for first signal using Gen4 waveforms.

```
With TheHdw.MWMultiToneSource("RF_PIN").Signals(SigName).Tones(1)  
    .SampleRate = sampleRateOne  
    .SampleSize = sampleSzOne  
    .Type = tLMWSignalTypeIF  
    .ReverseSpectrum = True  
    .Frequency = rf_freq  
    .CrestFactor = 3.1  
    .Amplitude = rf_level  
    .WaveDefinitionName = modsrc_segname
```

End With

' Set up parameters for second signal using Gen4 waveforms

```
With TheHdw.MWMultiToneSource("RF_PIN").Signals(SigName).Tones(2)  
    .SampleRate = sampleRateTwo  
    .SampleSize = sampleSzTwo  
    .Type = tLMWMTSignalTypeIF  
    .ReverseSpectrum = True  
    .Frequency = rf_freqI  
    .CrestFactor = 3.2  
    .Amplitude = rf_levelI  
    .WaveDefinitionName = modsrc_segnameI
```

End With

' Turn on modulation for both signals

```
TheHdw.MWMultiToneSource("RF_PIN").Tones(1).Modulated = True
```

```
TheHdw.MWMultiToneSource("RF_PIN").Tones(2).Modulated = True
```

```
' Add site loop here if needed
' Set up and connect the MWMultiToneSource
TheHdw.MWMultiToneSource.Pins("RF_PIN").Signals(SigName).LoadSettingsConnect

TheHdw.Wait (0.005)

'Start the signal from VBT or run a pattern to start the signal
TheHdw.MWMultiToneSource("RF_PIN").Signals(SigName).Start (tlStartImmediately)

.
.
.
.

' Disconnect the MWMultiToneSource
TheHdw.MWMultiToneSource.Pins("RF_PIN").Disconnect

End Function
```

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UltraWave 12G MWMultiToneSource Programming : MWMultiToneSource Pattern Control

MWMultiToneSource Pattern Control

You can control an MWMultiToneSource instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming.

You create a pattern independently of the test program using the PatternTool or a text editor.

Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling the instrument during pattern execution.

The following topics are available:

- MWMultiToneSource Pattern Microcodes
- MWMultiToneSource ASCII Pattern Syntax

☒ **Note:**

To use pattern microcodes to control an MWMultiToneSource on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

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UltraWave 12G MWMultiToneSource Programming : MWMultiToneSource Pattern Control :
MWMultiToneSource Pattern Microcodes

MWMultiToneSource Pattern Microcodes

A microcode controls a single function, such a starting or stopping the sourcing of a signal. Extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is sourced. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start generating a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used to issue commands to an MWMultiToneSource instrument from a pattern.

MWMultiToneSource Pattern Microcodes

Microcode	Functional Description
Resync	Forces phase alignment of an MWMultiToneSource instrument clock with the HSD T0 clock. Resync is also required to align MWModulatedSources to each other (if time alignment is necessary). See Use of Resync.
Start <SignalName>	The instrument immediately (on the next available logical analog clock) starts sourcing the named signal in SMEM and repeats it continuously until a Stop microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the

	default signal is used (see TheHdw.MWMultiToneSource.Pins.Signals.DefaultSignal).
StartE <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the named signal in SMEM and repeats it continuously until a STOP microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWMultiToneSource.Pins.Signals.DefaultSignal). If the MWMultiToneSource is being used to source two sample based Signals, then each AWG has an independent end ("E"). The next sample based waveform will start when its AWG completes the waveform it is currently sourcing. When the two AWGs are operating with a non 1:1 waveform length, the next sample based waveforms will start at different times.
Start1 <SignalName>	The instrument immediately starts sourcing (on the next available logical analog clock) the named signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWMultiToneSource.Pins.Signals.DefaultSignal).
Start1E <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the named signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWMultiToneSource.Pins.Signals.DefaultSignal). If the MWMultiToneSource is being used to source two sample based Signals, then each AWG has an independent end ("E"). The next sample based waveform will start when its AWG completes the waveform it is currently sourcing. When the two AWGs are operating with a non 1:1 waveform length, the next sample based waveforms will start at different times.

Stop	The instrument stops sourcing the currently sourced signal immediately. STOP does not impact a signal that is running once.
StopE	The instrument stops sourcing the currently sourced signal at the end of the signal. If the MWMultiToneSource is being used to source two sample based Signals, then each AWG has an independent end ("E"). Each AWG will stop when it completes the waveform it is currently sourcing. When the two AWGs are operating with a non 1:1 waveform length this means the stopping point of the sample based waveforms will occur at different times.
LoadSettingsConnect <SignalName>	Instructs an MWMultiToneSource to load the settings for the named capture signal and connects the hardware to the DUT. SignalName is required.
Disconnect	Instructs an MWMultiToneSource to reset all connections to their disconnected state.
Enable_Inst_Cond	Enables the instrument condition. The MWMultiToneSource asserts the instrument condition to the pattern generator at the completion of a capture. A MWMultiToneSource only supports this when using a sample based (for example, modulated) Signal on one or both of its paths. If a sample based Signal is being used on both paths and COND is being set on a specific sample, then if different sample rates are involved the COND is set by the AWG which reaches that sample first.
Disable_Inst_Cond	Disables an MWMultiToneSource from asserting the instrument condition.

Notes on MWMultiToneSource Microcodes

Note the following when using microcodes with an MWMultiToneSource:

- IG-XL has rules regarding the necessary spacing in vectors between microcodes on a specific channel. If you place microcodes too close together, IG-XL will return an error. General guidelines for spacing microcodes include the following:

- Resync: <1s

- LoadSettingsConnect: 200 - 400s (typical)

- Disconnect: <200s (typical)

- Start: <10s

- An MWMultiToneSource does not use PSets. It provides the equivalent capability using Signals programming in VBT.

- When using an UltraWave logical instrument (such as an MWMultiToneSource), any vector that contains a POP microcode must be fully defined across all defined logical instrument instances.

- IG-XL returns an error if you do not follow this guideline.

- Reapplying the same Signal to a given logical instrument instance results in the hardware NOPing the application. (This prevents glitches.)

For more information and an example, see [Using UltraWave 12G Instruments with POP](#).

Use of Resync

The Resync microcode in the Microwave subsystem performs additional tasks when compared to the Resync used with the BBAC and VHFAC instruments. The primary function of Resync for BBAC and VHFAC is to align the analog clock for the analog instrument with the digital clock supplying the timing reference for the pattern.

For the Microwave instruments, the Resync microcode also:

- Forces alignment of the internal AWG clocks which drive I and Q modulation AWGs

- Synchronizes the start of waveforms

The instantiation time associated with Resync is 1 s, which you must manage. Therefore, you must place a wait of at least 1 s between the Resync and following microcode.



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UltraWave 12G MWMultiToneSource Programming : MWMultiToneSource Pattern Control :
MWMultiToneSource ASCII Pattern Syntax

MWMultiToneSource ASCII Pattern Syntax

MWMultiToneSource pattern microcodes are instructions that you can place on an individual vector of a pattern. For that vector, a microcode instructs the MWMultiToneSource to perform a particular action, such as starting a new signal.

You can only use MWMultiToneSource microcodes with pins that the instruments statement has specified as MWMultiToneSource. For example:

```
instruments = {
    DUT_AIN:MWMultiToneSource;
}
```

For more information, see [Instruments Statement](#) in the Pattern Language section.

If a vector contains any MWMultiToneSource related microcode other than nop, that vector is placed in SRM.

The format for specifying an MWMultiToneSource microcode on a particular vector is:

(pin-list : MWMultiToneSource = microcode)

The following code shows an example of ASCII pattern microcode syntax for a MWMultiToneSource. The figure shows this pattern in PatternTool.

```
instruments = {
    Source:MWMultitoneSource 1;
}

vm_vector
vm_MWMultiToneSourceOnly_A ( $tset , Dig1)
{
    start_label MWMultiToneSourceOnly:
```

[illegible]

}

MWMultiToneSourceOnly.PAT (vm_MWMultiToneSourceOnly_A - VM)						
TSet	Command	Label	Vector	Source (MWMultitoneSource)	Digi	Comment
BasicSource		start_label MWMultiToneSourceOnly:	0		0	
BasicSource			1	Resync	0	
BasicSource	repeat 10		2		0	1uS wait time
BasicSource			3	LoadSettingsConnect GSM	0	
BasicSource	repeat 10000		4		0	1mS wait time
BasicSource			5	Start GSM	0	
BasicSource	repeat 100		6		0	10uS wait time
BasicSource			7	LoadSettingsConnect WCDMA	0	
BasicSource	repeat 10000		8		0	1mS wait time
BasicSource			9	Start WCDMA	0	
BasicSource	repeat 100		10		0	10uS wait time
BasicSource			11	Stop	0	
BasicSource	repeat 100		12		0	10uS wait time
BasicSource			13	Disconnect	0	
BasicSource	repeat 2000		14		0	200uS wait time
BasicSource			15		0	
BasicSource			16		0	

Vertical: 65 Vectors Vectors 0 to 16

MWMultiToneSource Pattern Microcodes Example

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UW12GMultitoneSrc_TOO

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UltraWave 12G MWMultiToneSource Theory of Operation

UltraWave 12G MWMultiToneSource Theory of Operation

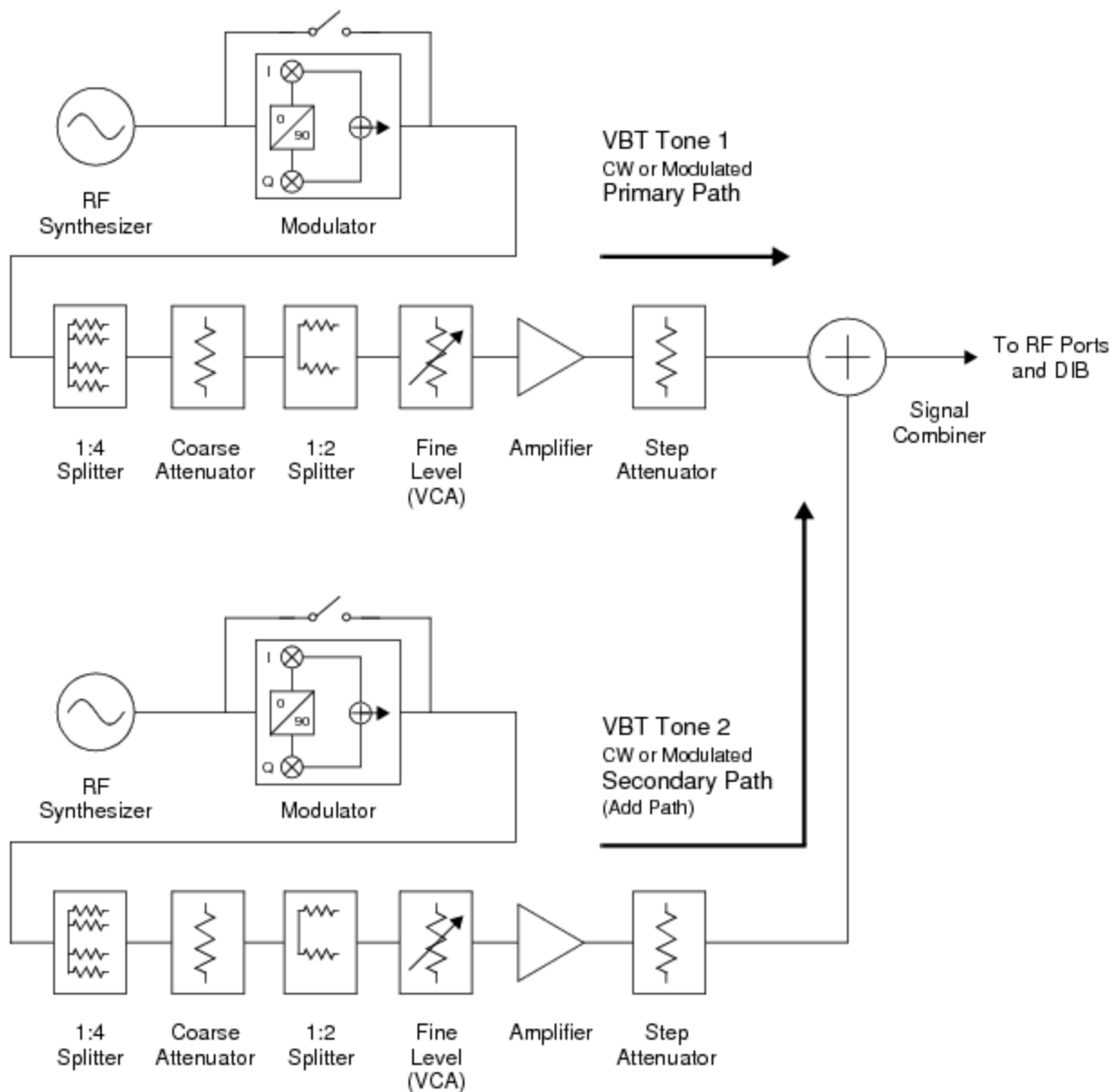
The MWMultiToneSource instrument produces a signal with two or more tones (signal frequencies) at programmable frequencies. The multitone signal is created by combining the outputs of two sources: each can be a MWSource or a MWModulatedSource. The following topics are available:

- [Basic Operation](#)
- [MWMultiToneSource Hardware Functions and Signal Flows](#)

For information on MWMultiToneSource instrument performance specifications and standards, see [UltraWave 12G Specifications](#).

Basic Operation

Each of the two source instruments in a MWMultiToneSource generates signals in the same way as an individual CW or modulated source. However, the outputs of the two sources are routed through a signal combiner, or add path, function. See the figure.



UltraWave 12G MWMultiToneSource Block Diagram

Note:

The signal path of a MWMultiToneSource has more than one separate attenuator and amplifier. All level control elements are combined in the blocks in the diagram.

The figure represents a CW source with a switched bypass path around the modulator. For both a CW and modulated signal, the signal path characteristics before the signal combiner are different for the two sources:

- The source signal routed directly to the signal combiner in a front end uses the primary path. This path has highest performance, including higher power output.
- The source signal routed out of the other front end uses the secondary path. This path has lower performance than the primary path, including reduced power output.

Specifications for the paths are included in the UltraWave Specification. For a diagram of the paths, see [MWMultiToneSource Hardware Functions and Signal Flows](#).

For a description of the operation of an individual CW source, see [CW Source Basic Operation](#) in the MWSource instrument manual.

For a description of the operation of an individual modulated source, see [Basic Operation](#) in the MWModulatedSource instrument manual.

Note:

The main RF input path to the MWReceiver is blocked on a front end used by the secondary signal for a MWMultiToneSource because the secondary source takes over the step attenuator. In this case, a MWReceiver connected on the front end must use the alternate path. See [TheHdw.MWReceiver.Pins.RFInputPath](#).

[MWMultiToneSource Hardware Functions and Signal Flows](#)

The figure shows the configuration of the UltraWave 12G subsystem and source signal flow for the operation and connection of a MWMultiToneSource adding two CW signals. Heavy lines represent source signal flows. The primary and secondary sources and paths are labeled.

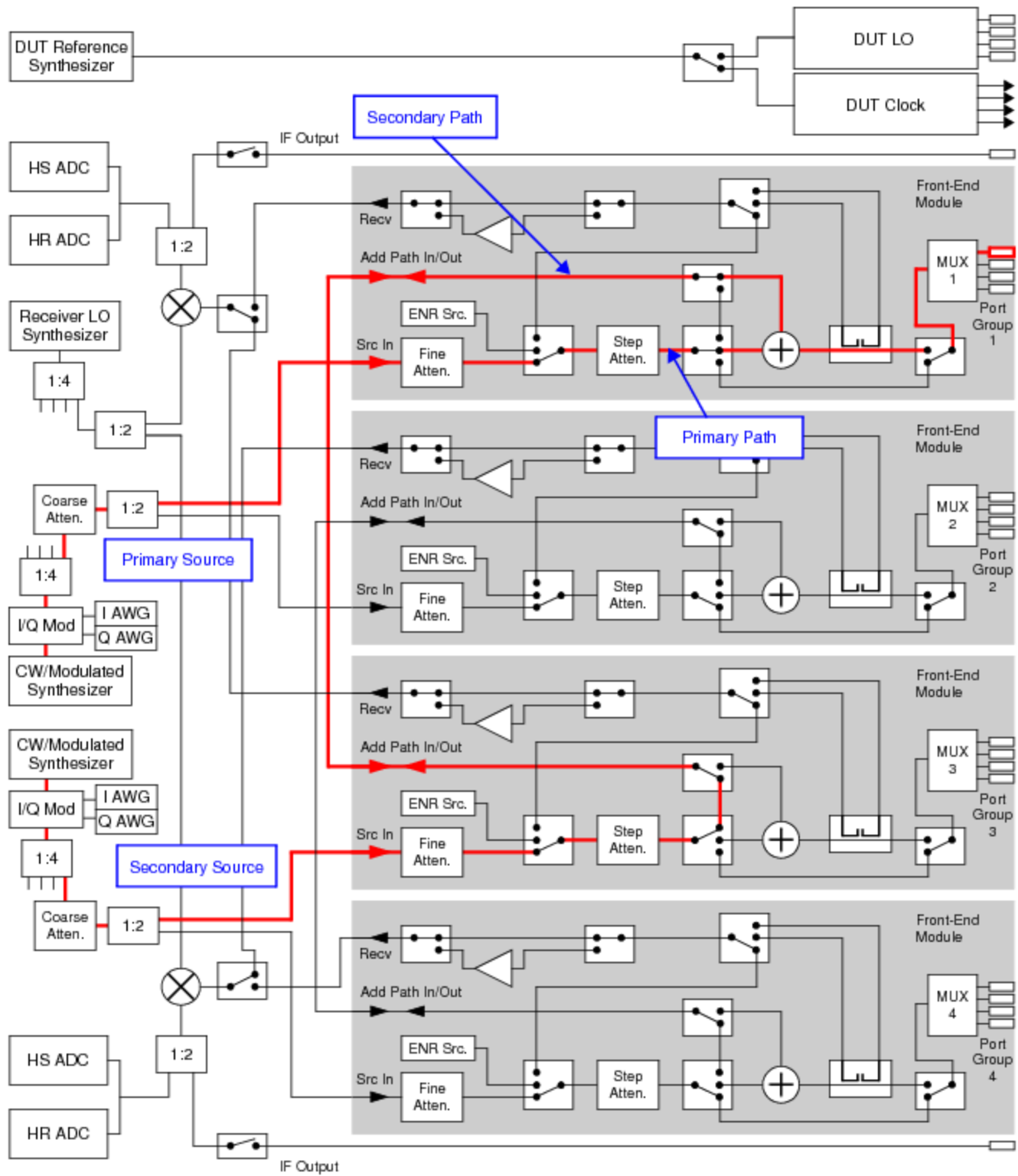
The source routed into the front end for port group 1 uses the primary path. The primary path is the front end with the port where the MWMultiToneSource pin is assigned. The source routed into the front end for port group 3 uses the secondary path.

The figure shows that the MWMultiToneSource can only use two combinations of front ends/port groups:

- The front ends for port group 1 and pin group 3
- The front ends for port group 2 and pin group 4

In both cases, either front end can function as the primary signal path, depending on the channel assignment for the MWMultiToneSource.

The two sources do not share the same RF synthesizer. The MWMultiToneSource uses two separate CW/modulated sources, with an RF synthesizer for each one. However, if there are MWMultiToneSources on more than one Measure board, they share RF Synthesizers.



MWMultiToneSource Hardware Functions and Signal Paths



